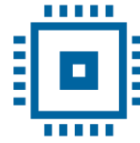




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Verification Project

Apartment Boiler Control

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1 THE DIGITAL DESIGN OF THE CIRCUIT.

Block Diagram

General Functionality Description of the DUT

DUT Interfaces

List of DUT Functionalities

Schematic Diagram

Waveforms

Register Table

1.1 BLOCK DIAGRAM

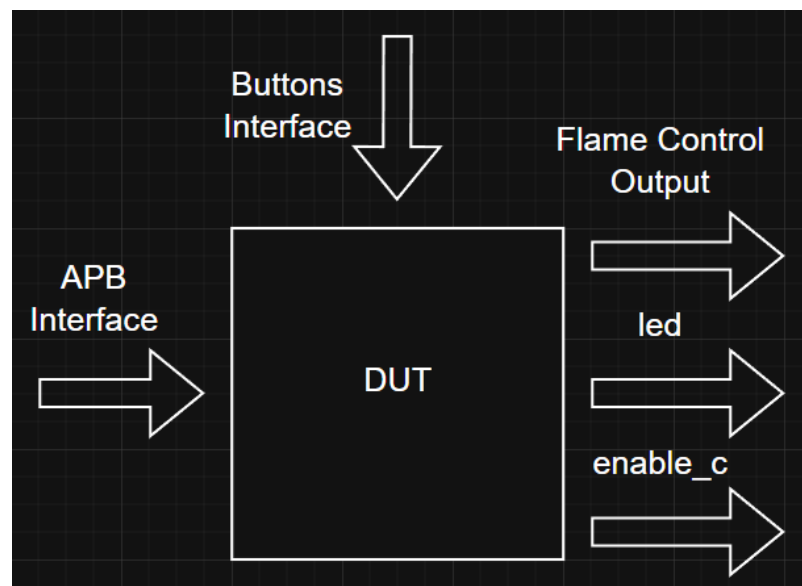


Figure 1. DUT - Apartment Boiler Control

1.2 GENERAL FUNCTIONALITY DESCRIPTION OF THE DUT

The DUT enables control over an apartment boiler, specifically for igniting the flame and maintaining it for a well-defined period, depending on the boiler's configuration. The

functionalities of the boiler are restricted by the manufacturer, while its usage is entirely dependent on the user.

1.3 DUT INTERFACES

Table 1. Signals in the Configuration Interface – APB

Data Transmission Interface			
Signal Name	Bit Width	Direction	Description
pclk	1	Input	Synchronization signal
paddr	2	Input	Address being accessed
pwrite	1	Input	Indicates write when HIGH and read when LOW
psel	1	Input	Slave device selection
penable	1	Input	Indicates the next transfer cycles
pdata	8	Input	Contains the data to be written
pready	1	Output	Acknowledgment signal
prdata	8	Output	Contains the data to be read

Table 2. Signals in the Button Interface

Configuration Interface			
Signal Name	Bit Width	Direction	Description
button_mod_econ	1	Input	Selects economy mode
button_mod_comf	1	Input	Selects comfort mode
button_mod_thermo	1	Input	Selects thermostat mode
button_child_prot	1	Input	Selects child

			protection mode
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Table 3. Output Interface Signals

Result Display Interface			
flame	1	Output	Ignition signal
led	1	Output	Power indicator — active in all states except s0
enable_c	1	Output	Counter enable — active only in econ and comf states

1.4 LIST OF DUT FUNCTIONALITIES

- The system can be turned on/off via a button.
- After shutting down, the system resets on startup, beginning operation from state s1.
- A register in the DUT can be programmed to activate functionalities (e.g., econ/comf) via an APB interface.
- The DUT allows selecting one of the modes using buttons through a button interface.
- The DUT controls the duration for which the flame remains lit, depending on the selected mode.
- An LED indicates the boiler's power status.
- Provides an economy mode option.
- Provides a comfort mode option.
- Child protection mode can be activated, preventing other buttons from functioning.

The DUT starts in the mode selection state, s1. Assuming register 0x00 is configured as 1111, enabling all DUT functionalities, button presses are checked. If the economy mode button is pressed, the DUT transitions to the state handling economy mode functionalities. Pressing any other mode button will deactivate the logical 1 state of the economy mode button and activate a different functional mode.

For example, if the DUT is in economy mode and the comfort mode button is pressed, the system transitions to comfort mode. Pressing buttons while another mode is active will switch the system from one state to another. If the child protection mode button is pressed, all other buttons will be disabled until the protection button is set to logical 0.

Handling each functional state involves maintaining the flame for a specific duration, which corresponds to a certain number of clock cycles.

Other DUT Signals

- **counter** – Internal signal, represents a counter used to maintain the "flame" signal for a specific number of clock cycles.
- **enable_c** – Output signal, used to activate the counter.
- **data_in** – Internal signal, used for loading data into the counter.
- **finish** – Internal signal, indicates the completion of the counter's counting process.

Encountered Issues:

Using a register (**func_reg**) to limit DUT operation by restricting access to its states. This was implemented using a logical **AND** check on the corresponding bit from **func_reg** for each state. Implementing the **child protection** function (button press protection) – Similar to functionality restriction, an **AND** check is used when transitioning between states. **child_protection** must be set to logical **0** to allow state transitions via button presses.

1.5 WAVEFORMS

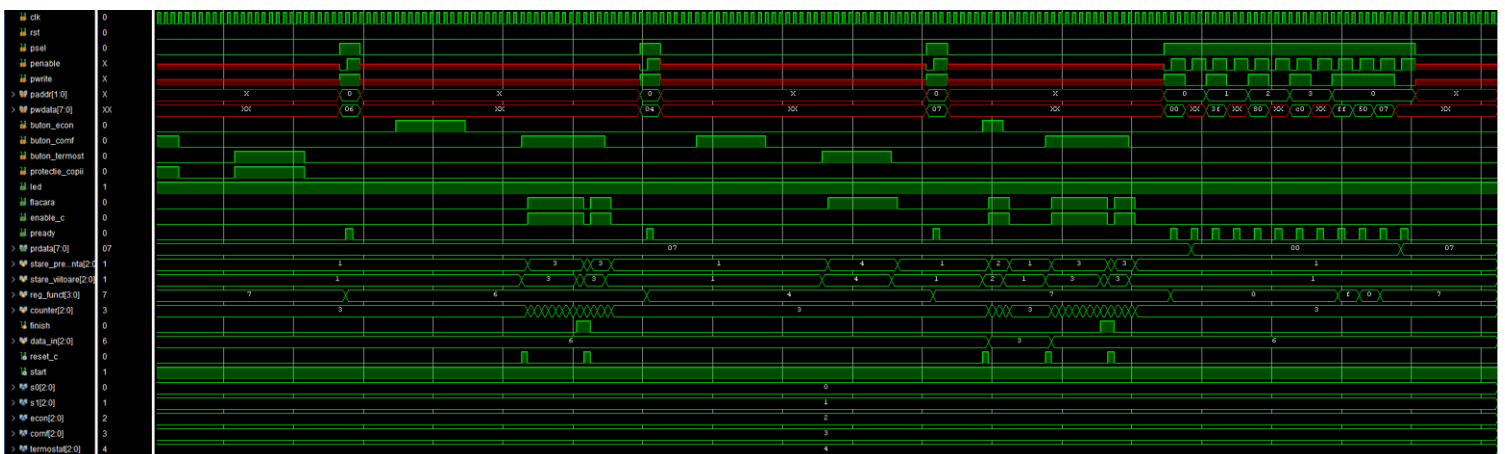


Figure 2. Simulation of the First Version of the DUT in Vivado

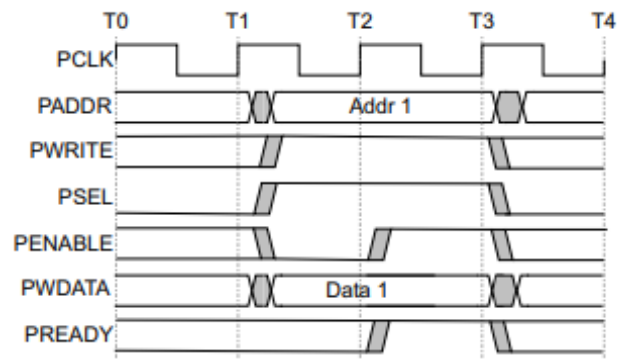


Figure 3-1 Write transfer with no wait states

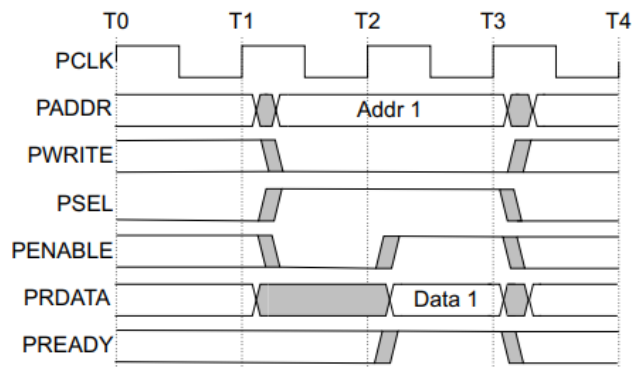


Figure 3-4 Read transfer with no wait states

Figures 3, 4. Waveforms for APB Transfer

1.6 REGISTER TABLE

Table 4. Register Table Accessible by the DUT

Register Name	Access Address	Register Structure	Register Field Description																
CFG	0x00	<table><tr><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td></td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>	7	6	5	4	3	2	1	0									bit0- ECON_MODE — enables economy mode bit1- COMF_MODE — enables comfort mode bit2- TERMOST_MODE — enables thermostat mode bit3- RESERVED: not used in current implementation bit 4:7- DON'T CARE(RESERVED)
7	6	5	4	3	2	1	0												

2. VERIFICATION OF THE DIGITAL CIRCUIT DESIGN

Verification Environment Architecture

Checklist of Elements to Verify

Functional Coverage Elements

Tests

Obtained Results

2.1 VERIFICATION ENVIRONMENT ARCHITECTURE

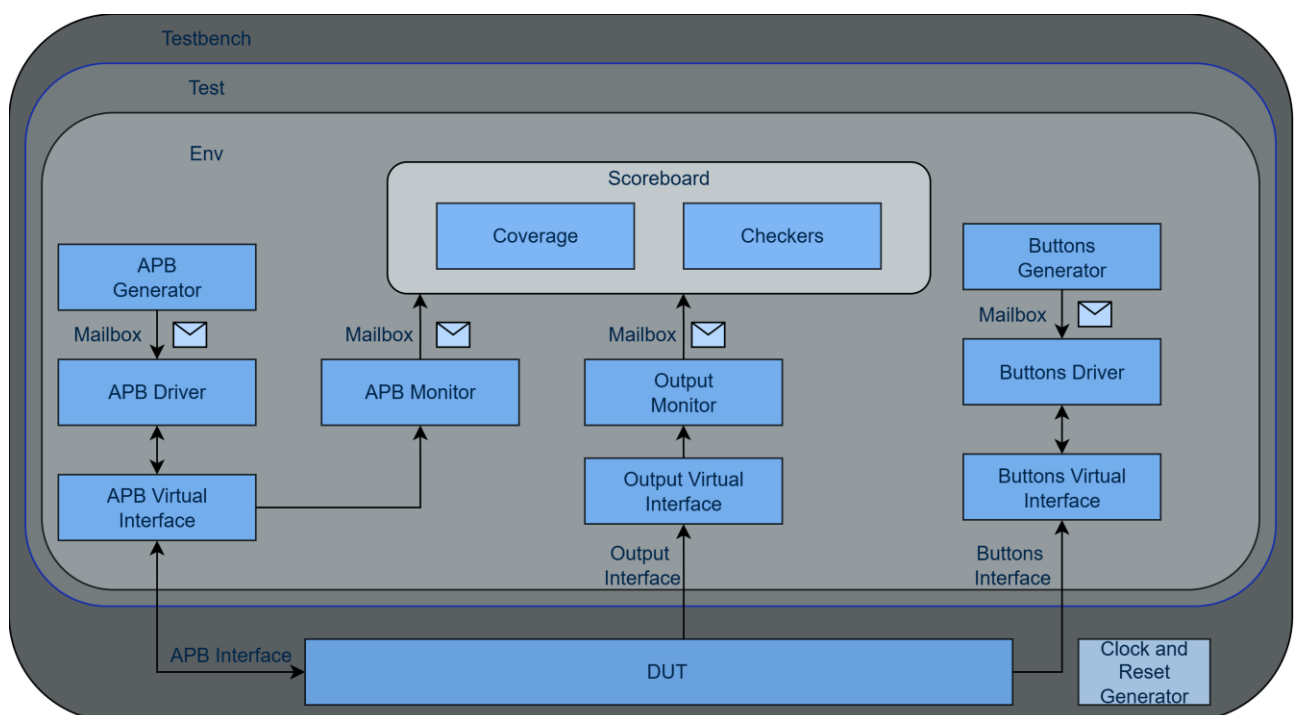


Figure 3. Overview of the components necessary for sending stimuli to the DUT via the APB interface

2.1.1 Driver for Button Interface

The Buttons Driver receives button transactions from the mailbox shared with the Buttons Generator. For each transaction, it waits for the specified delay (in clock cycles), then asserts the corresponding button signals (`button_mod_econ`, `button_mod_comf`, `button_mod_thermo`, `button_child_prot`) on the buttons interface. The signals are held for a configurable number of clock cycles (the hold field), after which all buttons are deasserted. This level-sensitive behavior matches the DUT's requirement that buttons must be held for multiple cycles to allow state transitions and counter operation.

2.1.2 APB Driver

The APB Driver receives transactions from the APB Generator via a mailbox. It drives the APB protocol signals (`paddr`, `pwrite`, `psel`, `penable`, `pdata`) on the APB interface following the

standard APB handshake: setup phase (psel=1, penable=0), followed by access phase (penable=1), waiting for pready acknowledgment from the DUT. It supports both write and read operations.

APB Monitor

2.1.3 Output Interface Monitor

The Output Monitor continuously observes the flame signal (flame_on) on the output interface. Each clock cycle, it captures the signal value into an output_transaction, sends it to the scoreboard mailbox, and samples the output coverage collector.

2.1.4 APB Interface Transaction

This class contains attributes marked as "rand," meaning they will receive random values. Used to represent APB transactions in simulations.

2.1.5 Output Interface Transaction

The Output Transaction class contains a single random field: unit_on (boiler active), representing the flame signal state (1 = flame on, 0 = flame off). It is used by the Output Monitor to capture the current state of the DUT's flame output each clock cycle. The transaction also includes a do_copy() method for deep copying.

2.2 FUNCTIONAL COVERAGE ELEMENTS

Table 5. APB Coverage Group (apb_transaction_cg):

Cover Point	Description
pwrite_cp	Write/read transaction types
paddr_cp	All 4 register addresses (0–3)
data_cp	Data value ranges: 4 equal bins [0:255], plus bins for 0x00 and 0xFF

Table 6. APB Coverage Group (apb_transaction_cg):

Cover Point	Description
unit_on_cp	Flame signal states (0=off, 1=on)

Table 7. Buttons Coverage Group (transaction_cg):

Cover Point	Description
button_mod_econ_cp	Economy button states (0 and 1)
button_mod_comf_cp	Comfort button states (0 and 1)
button_mod_thermo_cp	Thermostat button states (0 and 1)
button_child_prot_cp	Child protection button states (0 and 1)

2.3 TESTS

Table 5. List of Implemented Tests for DUT Verification

Test File Name	Test Description
all_modes_test.sv	Comprehensive directed test: exercises all 3 operating modes (econ/comf/thermo), child protection blocking, register-based mode enable/disable, mode transitions, and full APB address/data coverage sweep.
onebutton_test.sv	Random single-button test: generates random button presses with child protection disabled, verifying FSM transitions.
random_test.sv	Basic random test: generates random button transactions after configuring func_reg.

2.4 OBTAINED RESULTS

After running all_modes_test.sv, the following coverage results were obtained:

- APB Interface: paddr = 100%, data = 100%, pwrite = 100%
- Output Interface: flame = 100%
- Buttons Interface: econ = 100%, comf = 100%, thermo = 100%, child_prot = 100%

All 10 directed tests passed, verifying: economy mode (counter target=3), comfort mode (counter target=6), thermostat mode (no counter), child protection blocking on all 3 modes, register-based mode disable, and mode-to-mode transitions.